Title of Project Proposal (15% of Project Mark)

Course: Data 602 Statistical Data Analysis
Group Members: [List names & UCIDs]

1. Dawson Conlon: 30310916

2. Ricky Doucette: 30095675

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Due Date: Friday, September 25 @ 11:59 pm

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1. Purpose:

1. **Domain:**

   E-commerce: The buying and selling of goods on the internet via computers [3].

2. **Investigation Goal:**

   Differentiate what makes a VIP, Premium and consumer customer, based on how they interact with the store and the demographic of said customers.

3. **Practical Implications:**

   This allows us to understand who the companies deem a high value customer, how to keep them satisfied, and how to attract more high value customers so the company can increase profit.

4. **Population(s) of Interest:**

   We are studying online consumers from the store (VIP, Premium, Consumer).

Customers purchasing from the theoretical online store who have been categorized based on how they have interacted with the store (VIP, Premium, Consumer) are the target population.

5. **Variables of Interest:**

   Associated population variable: Customer segment, age, gender, customer country, return status, customer rating, product cost, customer lifetime value, customer order count.

1.6 Data Collection Method:

Synthetically generated data for a hypothetical E-commerce store that was generated over the course of 4 years.

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2. Data:

1. **Source of Data:**

   E-Commerce Sales Analytics Dataset from Kaggle: link.

2. **Permission/Accessibility:**

   Publicly available.

3. **Time Frame of Data Collection**

   The dataset covers transactions from 2021 – 2025.

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3. Topic(s) to Investigate:

1. **Focus of Statistical Investigation:**

Questions to be investigated

Is there a significant difference in mean lifetime value between customer segments?

Is the mean lifetime value of higher priority customers significantly greater than those of lower priority customers?

Is there any relationship between age or gender with customer segment?

What is the correlation between customer segment, lifetime value, and order count?

2. **Background/Context:**

   This is an E-commerce store that sells a wide variety of different items i.e.) electronics, automotive, sports goods, ETC. They can ship internationally so they reach many customers of different demographics.

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4. **Statistical Methods**

   95% confidence intervals for customer rating, customer age, customer order count, and customer lifetime value based on customer segment.

   Hypothesis test on the difference in mean customer lifetime value based on customer segment.

   Sampling. Samples will be taken from the population dataset as the original data has 150,000 rows.

Assumptions:

Data is normally distributed.

Data points are independent from one another.

How will we check Assumptions?

Normal probability plot to check if there is a normal distribution.

Using built in r functions to take random samples to confirm independency.

Planned Methods:

Descriptive statistics (mean, median, mode, ETC)

Confidence intervals

Hypothesis test

Testing normality

Simple random sampling

Bootstrapping – If needed, based on the normality of the distribution

Visualizations planned to be used on exploratory data analysis

Histograms, boxplots, bar charts, and heatmaps.

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5. References

Dataset: https://www.kaggle.com/datasets/datascikhan/e-commerce-sales-and-customer-analytics/data?select=order_items.csv

Data 602 Notes: https://d2l.ucalgary.ca/d2l/home/769643

[3] What is Ecommerce: https://www.netsuite.com/portal/resource/articles/ecommerce/ecommerce.shtml